

Beyond Correlation: A Unified Framework for Stable Predictive Modeling in Financial Time Series Using Causal Inference and Distributional Robustness

This report outlines a comprehensive, theoretically grounded, and algorithmically detailed research framework for stability selection tailored for financial time series. The objective is to establish a foundational baseline for future empirical studies by unifying methodological approaches from resampling theory, causal inference, and distributionally robust optimization. While empirical validation is outside the scope of this work, the framework provides clear specifications for implementation, enabling rigorous data-driven research. The analysis synthesizes recent academic literature to propose a multi-stage methodology that moves beyond traditional correlation-based feature selection towards the construction of more robust, stable, and interpretable predictive models capable of withstanding the inherent non-stationarity and uncertainty of financial markets.

Theoretical Underpinnings: Integrating Resampling, Causality, and Distributional Robustness

A robust framework for stability selection in financial time series necessitates a synthesis of three distinct but complementary theoretical domains: resampling methods for statistical inference on dependent data, causal inference for identifying true drivers rather than spurious associations, and distributionally robust optimization for formalizing and ensuring out-of-sample performance guarantees. This integration creates a holistic methodology where each component addresses a specific weakness of the others, leading to a more resilient modeling paradigm. The core idea is to use resampling not just as a tool for estimating variability, but as a mechanism to probe model stability; to leverage causality not merely as an interpretability layer, but as a guiding principle for feature construction; and to frame the final model not as a single point estimate, but as a solution to a robust optimization problem designed to hedge against distributional shifts.

The foundation of stability selection is resampling, which Meinshausen and Bühlmann introduced to reduce false positives in high-dimensional variable selection [95](#) [177](#). The bootstrap is a versatile technique for estimating the sampling distribution of a statistic by repeatedly resampling from the original dataset [2](#) [66](#). For independent and identically distributed (i.i.d.) data, simple random sampling suffices. However, financial time series violate the i.i.d. assumption due to serial dependence, such as autocorrelation and volatility clustering [68](#) [193](#). Applying standard bootstrap methods would destroy these critical temporal dependencies, leading to inconsistent estimates and invalid inference [156](#). Consequently, specialized bootstrap methods for time series have been developed, primarily falling into two categories: block-based and model-based approaches [101](#). Block bootstrap techniques, such as the Moving Block Bootstrap (MBB), preserve local dependence by resampling contiguous blocks of observations, thereby maintaining the order and correlation structure within each block [43](#) [103](#). Model-based approaches, like the sieve bootstrap, first fit a parametric model (e.g., an autoregressive model) to the data, then resample the residuals and re-simulate the time series, allowing the model to capture the underlying data-generating process [100](#)[126](#). These resampling mechanisms form the engine of stability selection, providing the diverse subsamples needed to assess the consistency of feature selection across different data fragments.

While resampling addresses the statistical challenge of dependency, it operates on the given set of features and does not inherently distinguish between causal predictors and statistically correlated noise. In finance, confounding variables are ubiquitous, and selecting features based purely on correlation risks identifying spurious relationships that may not hold under different market conditions [8](#) [92](#). This is where causal inference offers a powerful complement. The Structural Causal Model (SCM) provides a formal language for describing causal relationships through a directed acyclic graph (DAG) and functional equations, moving the focus from joint probability distributions to functional dependencies [14](#) [45](#). Causal feature selection aims to identify the direct causes of a response variable, which are expected to be more stable across environments because they reflect the underlying structural invariance of the system [15](#) [76](#). By integrating causal principles, the feature pool for stability selection can be refined to include only those variables that plausibly drive the outcome, significantly reducing the risk of selecting features based on transient correlations. Techniques have been developed to adapt causal discovery algorithms, such as the Fast Causal Inference (FCI) algorithm, for time series data to infer effective connectivity networks [116](#)[149](#).

Finally, even if a stable set of causal features is identified, financial time series are characterized by non-stationarity and regime shifts, meaning the statistical properties of the data can change over time [12](#)[127](#). A model optimized solely on historical data may

perform poorly when faced with a new regime. Distributionally Robust Optimization (DRO) provides a principled framework to address this issue [162163](#). Instead of optimizing for performance on the empirical distribution of the training data, DRO optimizes for the worst-case expected performance over a set of plausible distributions, known as an ambiguity set, centered around the empirical one [72](#) [183](#). This approach explicitly hedges against uncertainty in the data-generating process. The Wasserstein metric has become a popular choice for defining the ambiguity set due to its strong finite-sample statistical guarantees, tractable reformulations for many problems, and intuitive interpretation [97](#) [125222](#). There is also a deep conceptual link between DRO and stability; notions of minimax optimal estimation under distribution shift are closely related to stability criteria [96](#), and paradigms like Invariant Risk Minimization (IRM) can be formulated as specific instances of DRO, highlighting a shared goal of learning invariant, causal structures [7](#) [9](#) [175](#).

Algorithmic Specification of the Resampling Engine for Time Series

The resampling engine is the operational heart of any stability selection framework, responsible for generating the diverse subsamples that allow for the assessment of feature selection stability. For financial time series, the choice of resampling algorithm is critical, as it must respect the inherent temporal dependence structure to produce valid results. The primary challenge is to balance the preservation of local correlation with sufficient variability between resampled datasets. The literature presents two dominant families of adapted bootstrap methods: block-based and model-based, each with distinct algorithmic implementations and parameter choices [101102](#).

Block bootstrap methods operate by treating the time series as a sequence of overlapping or non-overlapping blocks of consecutive observations. The most common variant is the Moving Block Bootstrap (MBB), where blocks of a fixed length m are drawn with replacement from the original series, and adjacent blocks are concatenated to form a new time series of the same length [43](#) [103](#). The key algorithmic parameter for MBB is the block length m . A small m fails to capture the serial correlation, while a large m results in highly correlated and less diverse resamples [4](#) [69](#). Consequently, automatic block-length selection procedures are crucial for practical application. Politis and White (2004) developed a prominent method that estimates the optimal block length from the data's

autocorrelation structure [4](#) . Other approaches involve minimizing an estimator of the mean squared error of the statistic of interest [63](#) . The consistency of the MBB for non-stationary time series has been established under certain conditions, though the proofs can be complex [64](#) [221](#). Other block bootstrap variants include the Stationary Bootstrap, which uses a random block length to ensure stationarity in the resampled series, and the Circular Block Bootstrap, which treats the time series as circular to avoid edge effects [101](#).

Model-based bootstrap methods offer an alternative strategy by first approximating the data-generating process with a parametric model. The sieve bootstrap is a notable example in this class [126](#). Its algorithm proceeds in several steps: 1. **Model Fitting:** Fit a sequence of autoregressive (AR) models of increasing order $p(n)$ to the observed time series, where $p(n)$ grows with the sample size n . This "sieve" progressively approximates a general linear process [126](#). 2. **Residual Bootstrapping:** Estimate the model residuals and create a new set of bootstrap residuals by sampling with replacement from the original estimated residuals. 3. **Re-simulation:** Generate a new synthetic time series of the same length by simulating the AR process using the bootstrap residuals.

The sieve bootstrap is particularly well-suited for linear processes and can be computationally efficient [100](#). Another model-based approach is the AR-sieve bootstrap, which is considered best for linear data [100](#). More generally, model-based techniques can be applied to various models like SARIMA and TBATS, where resampling is performed on the model's innovations or residuals [3](#) .

The following table summarizes the key characteristics of these primary resampling algorithms.

Feature	Moving Block Bootstrap (MBB)	Sieve Bootstrap
Core Principle	Resamples contiguous blocks of data to preserve local dependence 103 .	Approximates the data-generating process with a parametric model (e.g., AR) and resamples the model's innovations 126 .
Key Parameter	Block length (m). Requires an automatic selection method 4 .	Model order (p), which should increase with sample size 126 .
Assumptions	Weak mixing conditions are often assumed for theoretical guarantees 157 . Consistency has been shown for non-stationary series 64 .	Assumes the underlying process can be well-approximated by a sequence of AR models 126 . Often assumes linearity 100 .
Advantages	Intuitive, model-free approach. Widely used in financial applications 3 179 .	Can be computationally efficient for linear processes. Provides a clear model-based interpretation.
Disadvantages	Sensitive to block length choice. May struggle with long-range dependence.	Relies on the validity of the specified parametric model. Less flexible than block bootstrap.

For the proposed unified framework, the choice of resampling algorithm should be treated as a configurable option. The Moving Block Bootstrap is a strong default candidate due to its flexibility and widespread adoption, especially when combined with a robust automatic block-length selection procedure [65](#) [67](#). However, the framework should also accommodate the sieve bootstrap for cases where the data exhibits a clear parametric structure. The practitioner must acknowledge that the theoretical properties of these methods, such as consistency, rely on assumptions about the data's dependence structure (e.g., β -mixing or α -mixing conditions) which may be violated in real-world financial data [157219](#).

Causal Feature Selection as a Guiding Principle for Model Construction

Integrating causal inference into the stability selection framework elevates the process from correlational feature identification to the discovery of potentially stable, causal drivers. This is particularly valuable in financial markets, where spurious correlations abound and features selected based on association alone may lack generalizability across different market regimes [8](#) [76](#). The goal is to shift the feature selection criterion from "which variables are associated with the outcome?" to "which variables are direct causes of the outcome?". This aligns perfectly with the objective of stability selection, which seeks to identify features that are consistently selected, as direct causal features are hypothesized to be more invariant and thus more likely to be stably selected [13](#).

The theoretical foundation for this approach is the Structural Causal Model (SCM), a framework formalized by Judea Pearl that uses directed graphs and structural equations to represent causal relationships [16](#) [180244](#). Within an SCM, the effect of external interventions can be simulated using the `do()` operator, distinguishing mere observation from active manipulation [245](#). Causal feature selection focuses on identifying the parent nodes of a target variable in the causal graph, as these represent the direct causes [15](#). Identifying these direct causal features enhances model stability because they correspond to the fundamental mechanisms driving the system, which are more likely to remain constant than indirect correlations that can arise from unobserved confounders [13](#) [76](#). Recent research has provided necessary and sufficient conditions for causal feature selection even in the presence of latent variables, further strengthening the theoretical basis [12](#) [248](#).

There are several ways to integrate causal principles into a stability selection framework. One approach is **causal pre-filtering**. Before applying the stability selection algorithm, a causal discovery phase is conducted on the full dataset to learn a local causal graph. Algorithms like the Fast Causal Inference (FCI) algorithm, which has been adapted for time series, can be used to infer conditional independence relations and orient edges to find potential causal links [116](#). The parent nodes of the target variable in this learned graph would then constitute the candidate feature set for the subsequent stability selection step. This dramatically reduces the dimensionality of the search space and ensures that the stability selection process is focused only on variables with a plausible direct causal connection to the outcome.

Another strategy involves using causal concepts for **validation and refinement**. After an initial set of features is selected using a standard stability selection procedure (e.g., based on Lasso), causal strength metrics can be applied to the selected features to differentiate between those with a direct causal effect and those selected due to indirect correlations [249](#). For instance, one could estimate the direct causal effect (DCE) of each selected feature and examine if there is a clear boundary between features with large DCEs (causal) and those with small DCEs (non-causal) [249](#). Features found to be primarily associated through indirect paths could be deprioritized or removed, leading to a more parsimonious and potentially more stable final model.

Furthermore, recent advances suggest a more deeply integrated approach where causal reasoning is woven into the resampling process itself. Work combining bagging with causal graphical model algorithms has shown promise in improving the stability of graphical model learning [98 99](#). In such a framework, each bootstrap sample would be used to learn a local causal model, and feature stability would be assessed based on the consistency of the causal graph structures or the selection of causal parents across the bootstrap samples. This represents a more sophisticated fusion of the two paradigms, moving beyond simple feature pre-filtering to an iterative process of learning stable causal structures from resampled data.

The following table contrasts the different levels of integration for causal inference within the stability selection framework.

Integration Level	Description	Pros	Cons
Causal Pre-filtering	Use a causal discovery algorithm on the full dataset to identify a pool of direct cause candidates before applying stability selection.	Reduces dimensionality, focuses search on plausible features, improves interpretability.	Computationally intensive; causal discovery algorithms may have high false positive/negative rates; relies on assumptions like faithfulness 75 .
Causal Validation	Apply a causal strength metric to features already selected by a standard stability selection method to validate their causal status.	Can be applied as a post-hoc step; leverages existing selection outputs.	Does not prevent the inclusion of non-causal features during the initial selection phase; may be difficult if the number of selected features is large.
Integrated Learning	Learn a local causal model on each bootstrap sample and aggregate the results to derive a stable causal structure.	Most theoretically aligned; directly assesses the stability of causal relationships, not just associations.	Highly complex to implement; requires a combination of resampling and causal discovery algorithms 98 99 .

For the proposed unified framework, a modular approach is recommended. The framework should include a dedicated module for causal feature identification. Initially, this could be implemented as a pre-filtering step to define the feature pool. As the field matures, this could evolve into a more integrated learning scheme. It is important to note that all causal inference methods rely on assumptions about the data-generating process, such as the absence of unmeasured confounders or the faithfulness assumption, which can be challenging to verify in practice [92](#) [147](#). Acknowledging these limitations is essential for the proper application of causal principles.

Distributionally Robust Optimization as a Formalism for Out-of-Sample Performance

Distributionally Robust Optimization (DRO) provides the third pillar of the unified framework, offering a formal and mathematically rigorous way to address the pervasive uncertainty and non-stationarity in financial time series. Standard machine learning approaches typically optimize a model's performance based on the empirical distribution of the training data, implicitly assuming that future data will follow the same distribution. This assumption is frequently violated in financial markets, where regime shifts, structural breaks, and changing volatility patterns are common [121](#)[193](#). DRO mitigates this risk by seeking a solution that performs well against a range of possible future distributions, effectively hedging against model misspecification and data uncertainty [36](#) [162](#).

The central concept in DRO is the ambiguity set, which is a collection of probability distributions that are deemed "close" to the empirical distribution derived from the observed data [17, 183]. The DRO problem is then formulated as a minimax optimization: minimize the worst-case expected loss over all distributions contained within this ambiguity set. Mathematically, this can be expressed as:

$$\min_{y \in Y} \max_{P \in \mathcal{A}} \mathbb{E}_P[\text{Loss}(y, \xi)]$$

where y is the decision or model parameter to be optimized, Y is the feasible set, P is a probability distribution in the ambiguity set $\mathcal{A}$, and ξ represents the uncertain data. The critical element of this formulation is the definition of the ambiguity set $\mathcal{A}$. The choice of the distance metric used to construct this set determines the nature of the robustness being achieved.

The Wasserstein metric has emerged as a particularly attractive tool for constructing ambiguity sets in data-driven DRO. Several papers highlight its benefits, including the fact that it can be defined using the Wasserstein distance to constrain the set of distributions considered [178, 182]. Modern measure concentration results guarantee that, under mild assumptions, the true underlying distribution lies within a Wasserstein ball of a certain radius around the empirical distribution with a high probability that can be quantified [97, 222]. This provides finite-sample, non-asymptotic guarantees, which is a significant advantage over asymptotic arguments. Furthermore, DRO problems formulated with Wasserstein ambiguity sets can often be reformulated into computationally tractable convex optimization problems, making them applicable to a wide range of machine learning tasks [125, 260]. The radius of the Wasserstein ball can be interpreted as a budget for the permissible distributional perturbation, linking the level of conservatism in the model to a concrete statistical quantity [168, 232].

There are deeper connections between DRO and the other components of the framework. Firstly, the concept of stability is intrinsically linked to distributional robustness. Research has shown that minimax optimal estimators under distribution shift are related to notions of stability, suggesting that DRO provides a formal mathematical language for what it means for a model to be "stable" [96]. Secondly, the paradigm of Invariant Risk Minimization (IRM), which aims to learn predictors based on invariant causal associations that hold across different environments, can be cast as a specific type of DRO problem [7, 9, 175]. This connection bridges the gap between the goal of learning stable causal features (from the causal inference pillar) and the formal machinery of DRO. By framing the final model as a DRO problem, the framework explicitly optimizes for the kind of stability that IRM aims to achieve.

Advanced DRO formulations further enrich the framework. Decision-dependent DRO allows the ambiguity set to depend on the decision variable y itself. This can be used to calibrate the level of robustness dynamically, balancing the desire for robustness against the potential loss in nominal performance [47](#) [169173](#). Group-wise DRO applies the robust optimization principle separately to predefined subgroups of the data (e.g., bull markets, bear markets), ensuring that the model is not only robust on average but also performs well within each specific regime [34](#). These advanced concepts provide additional tools for tailoring the robustness of the final predictive model to the specific needs of the financial application.

An Integrated Algorithmic Blueprint for Stable Predictive Modeling

Synthesizing the theoretical pillars of resampling, causality, and distributional robustness yields a comprehensive, multi-stage algorithmic blueprint for stability selection in financial time series. This framework is designed to be modular, allowing for the incorporation of different methods at each stage while maintaining a coherent overall structure. The goal is to transform a raw dataset into a robust, predictive model with formal guarantees on its stability and out-of-sample performance. The process can be broken down into three main stages: Data Preparation and Feature Generation, Resampling and Stable Selection, and finally, Robust Modeling via Distributional Optimization.

Stage 1: Data Preparation and Feature Generation This initial stage focuses on transforming the raw input data into a more informative feature space and, optionally, refining the feature pool using causal insights. 1. **Input:** A multivariate financial time series dataset, consisting of a target variable (e.g., future returns) and a set of candidate predictor variables. 2. **Temporal Representation Learning:** Raw time series are processed to extract richer representations. This can involve standard techniques like creating lagged variables, rolling window statistics (mean, volatility), and technical indicators. More advanced methods, such as shapelet-based transformations or signature kernels, can be employed to capture complex temporal motifs and nonlinear dynamics, providing the model with more powerful inputs [77](#) [124239](#). 3. **(Optional) Causal Pre-filtering:** To enhance stability and interpretability, a causal discovery algorithm suitable for time series (e.g., tsFCI) can be applied to the dataset [116](#). The output is a causal graph from which the parent nodes of the target variable are identified. These nodes form the

reduced set of candidate features for the next stage, effectively filtering out variables that are not direct causes [249](#).

Stage 2: Resampling and Stable Selection This core stage executes the stability selection procedure, using resampling to assess the consistency of feature selection across different data fragments.

- 1. Define Resampling Strategy:** Select a resampling algorithm appropriate for time series. The Moving Block Bootstrap (MBB) is a recommended default due to its model-free nature [101](#). If the MBB is chosen, an automatic block-length selection method (e.g., based on spectral density or cross-validation) must be specified to determine the block size m [4](#) [65](#).
- 2. Bootstrap Loop:** Execute the following steps B times, where B is a large number of iterations (e.g., 1000).
 - a. Draw a bootstrap sample $S^{(b)}$ of the same length as the original series using the chosen resampling method.
 - b. On the subsample $S^{(b)}$, train a base selection model. The choice of the base algorithm is a configurable parameter; options include penalized regression (e.g., Lasso), tree-based methods (e.g., Random Forests, Gradient Boosting), or other high-dimensional selection algorithms [94](#).
 - c. Record the set of features, $\hat{S}^{(b)}$, that are selected by the base model on this particular subsample.
- 3. Aggregation and Final Selection:** Aggregate the results from all B bootstrap iterations to derive a stable set of features.
 - a. Calculate the selection frequency for each candidate feature j as $\hat{\pi}_j = \frac{1}{B} \sum_{b=1}^B I(j \in \hat{S}^{(b)})$, where $I(\cdot)$ is the indicator function [95](#).
 - b. Derive the final stable feature set, $\hat{S} = \{j \mid \hat{\pi}_j \geq \pi_{thr}\}$, using one of two common aggregation strategies. The **threshold-based approach** selects all features whose frequency exceeds a predefined threshold π_{thr} . The **rank-based approach** selects the top q features with the highest selection frequencies [95](#).

Theoretical analysis suggests the rank-based approach may be more robust to certain types of contamination [95](#).

Stage 3: Robust Modeling via Distributional Optimization This final stage takes the stable set of features and builds a predictive model that is formally robust to distributional uncertainty.

- 1. Train Robust Model:** Instead of simply fitting a standard regression or classification model to the full dataset using the features in $\hat{S}_{stable}$, formulate the problem as a Distributionally Robust Optimization task.
- 2. Define Ambiguity Set:** Construct a Wasserstein ambiguity set around the empirical joint distribution of the selected features and the response variable. The radius of this set, ϵ , must be calibrated from the data. Methods exist to choose ϵ such that the true distribution is contained within the set with a high probability, providing finite-sample guarantees [166](#)[168](#)[233](#).
- 3. Solve DRO Problem:** Solve the resulting minimax optimization problem. The objective is typically to minimize the worst-case expected loss (e.g., squared error for regression, log-loss for classification) over the Wasserstein

ambiguity set. This can often be reformulated into a tractable convex optimization problem [125178](#). The solution yields a robust model parameter (e.g., regression coefficients) and provides a prediction that is guaranteed to have good performance across a range of plausible future scenarios.

This integrated blueprint provides a complete, theoretically motivated workflow. It begins by creating informative features, uses resampling to filter for stable predictors, and concludes by building a model that is explicitly designed to be robust to the uncertainties inherent in financial time series. Each stage is supported by a rich body of academic literature, making it a solid foundation for future empirical investigation.

Synthesis, Limitations, and Directions for Future Research

This report has presented a unified theoretical and algorithmic framework for stability selection in financial time series, integrating resampling theory, causal inference, and distributionally robust optimization. The proposed framework moves beyond traditional methods by addressing three critical weaknesses of conventional modeling pipelines. First, it acknowledges the non-i.i.d. nature of financial data by employing specialized resampling techniques like the Moving Block Bootstrap. Second, it seeks to mitigate the problem of spurious correlations prevalent in finance by incorporating causal principles to guide feature selection towards more stable, direct causes. Third, it confronts the challenge of non-stationarity and regime shifts by framing the final model as a distributionally robust optimization problem, which provides formal guarantees on out-of-sample performance. Together, these components create a cohesive methodology for constructing predictive models that are not only accurate on historical data but are also fundamentally designed for stability and reliability in dynamic environments.

Despite its theoretical coherence, the proposed framework is subject to several limitations and presents significant challenges for future research. A primary concern is **computational complexity**. While DRO problems with Wasserstein ambiguity sets are often reformulatable into tractable forms, the dimensionality of the problem can grow rapidly with the number of features and samples [131133](#). Similarly, causal discovery on multivariate time series can be computationally demanding, and running these algorithms within a bootstrap loop for stability selection could be prohibitive for very large datasets. Developing more efficient algorithms for each stage of the framework is a critical area for future work.

A second major limitation is the reliance on **assumptions and hyperparameter tuning**. The consistency of block bootstrap methods depends on weak dependence conditions like mixing, which may not hold perfectly in financial data [64 219](#). Causal discovery algorithms rely on assumptions such as the faithfulness condition, which posits that all statistical dependencies are consequences of the causal structure, an assumption that is difficult to verify and may be violated [75](#). Furthermore, the framework introduces several hyperparameters whose optimal values are not known a priori: the number of bootstrap samples B , the block length for MBB, the selection threshold π_thr , and the radius of the Wasserstein ambiguity set. The sensitivity of the final model to these choices requires careful investigation, and automated methods for calibration are needed.

The **integration of the three pillars** also poses a significant engineering and methodological challenge. While the theoretical connections between bootstrapping, causality, and DRO are becoming clearer, developing a seamless, end-to-end software implementation that integrates these disparate components is a non-trivial task. The interfaces between the causal pre-filtering module, the resampling engine, and the DRO solver need to be carefully designed to ensure compatibility and efficiency. Moreover, the theoretical guarantees of each component (e.g., consistency of the MBB, soundness of the causal discovery algorithm, optimality of the DRO solution) are typically derived under separate, often idealized, assumptions. Proving composite guarantees for the entire integrated framework remains a profound open question.

Future research should focus on several key directions. Empirical validation of the framework's superiority over traditional methods on real financial datasets is paramount. This includes benchmarking its performance across different market regimes and asset classes. Research into scalable algorithms, particularly for causal discovery and DRO, is essential for practical adoption. Investigating adaptive methods for hyperparameter selection, perhaps using meta-learning or Bayesian optimization, could help automate parts of the framework. Finally, extending the framework to handle streaming data and incorporate test-time adaptation techniques would make it even more relevant for real-world financial applications, where models must continuously adapt to new information without catastrophic forgetting [253254](#). By systematically addressing these limitations, the proposed framework can evolve from a theoretical blueprint into a powerful and practical tool for building robust predictive models in finance.

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